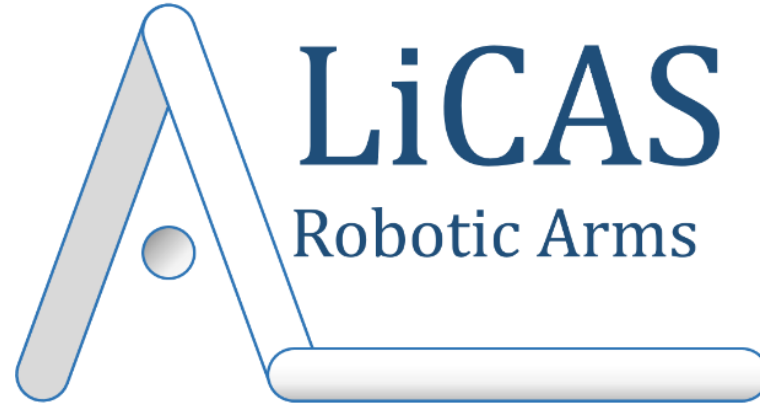


LiCAS A1

Anthropomorphic Dual Arm System

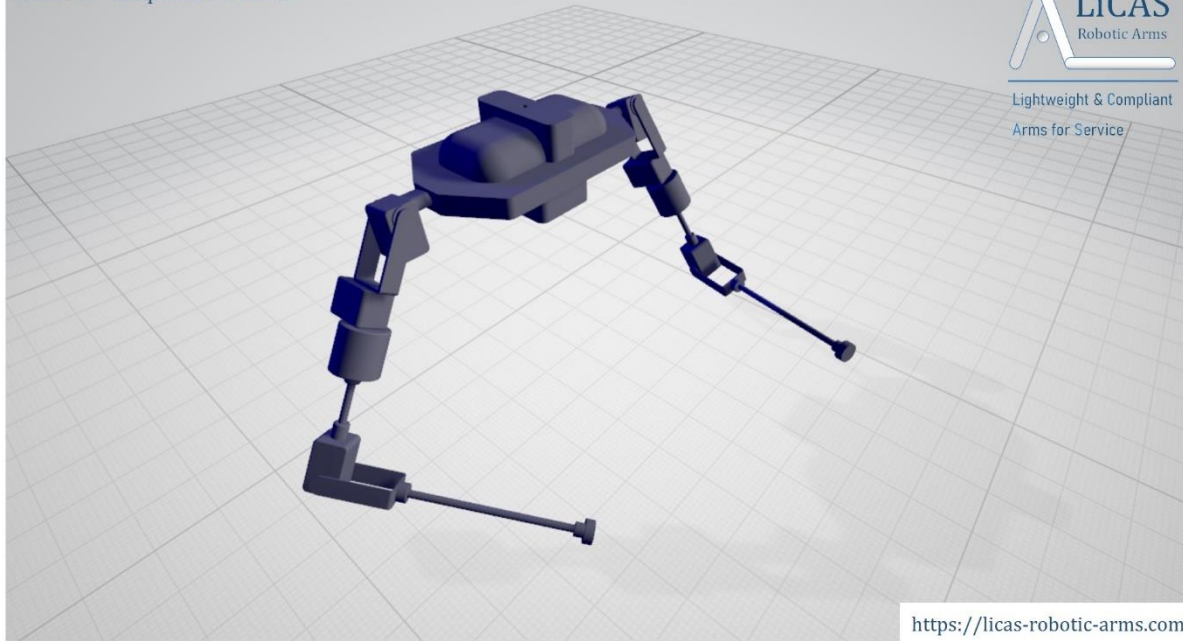
Simplified 3D Model Mass and Inertia Parameters

<https://licas-robotic-arms.com>

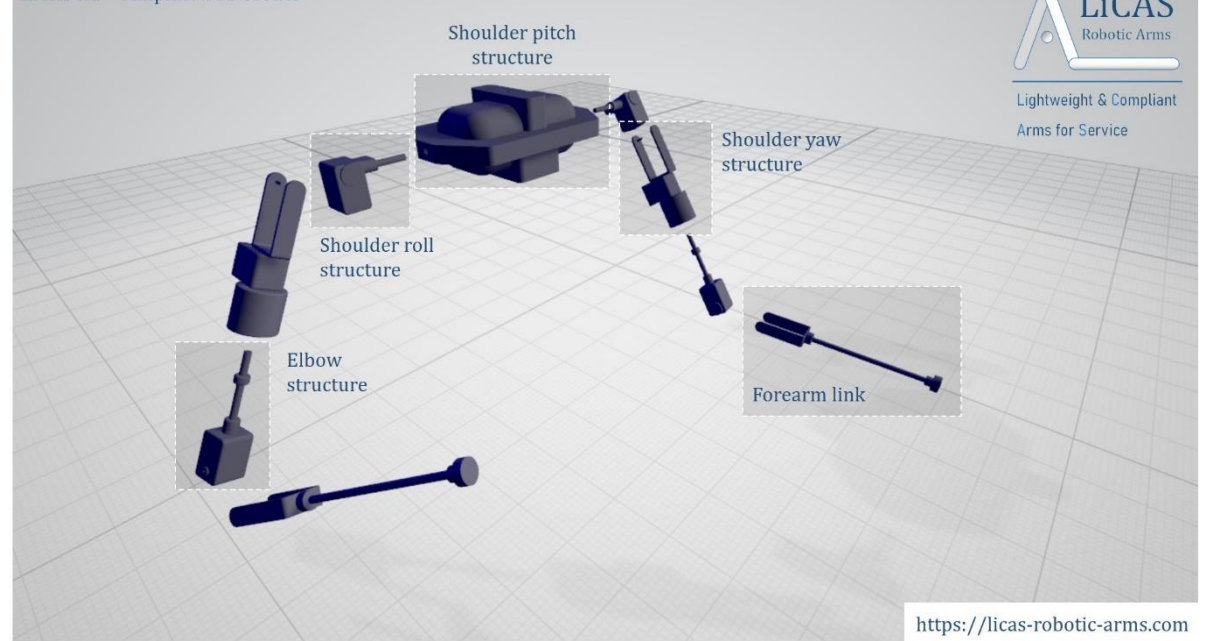


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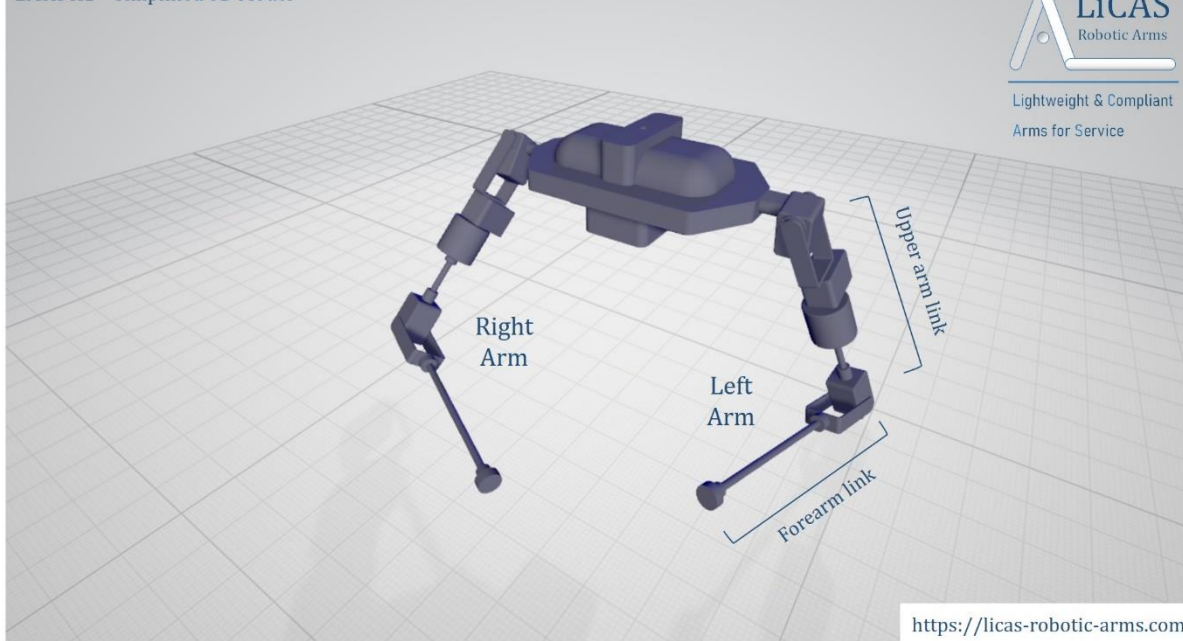
LiCAS A1 – Simplified 3D Model



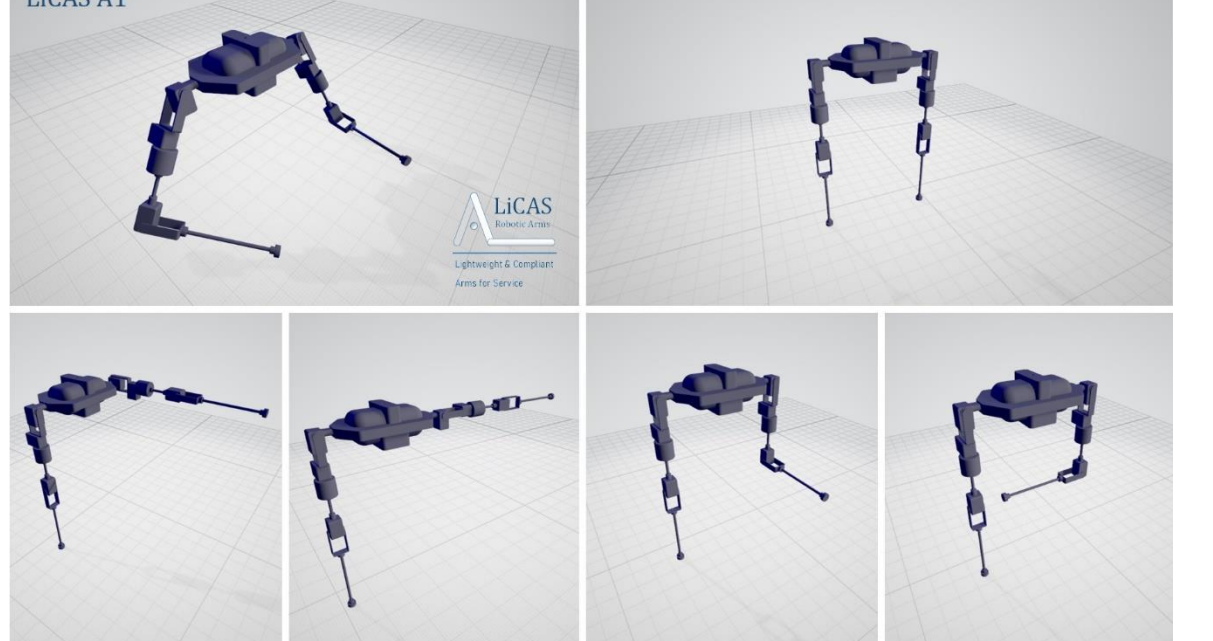
LiCAS A1 – Simplified 3D Model



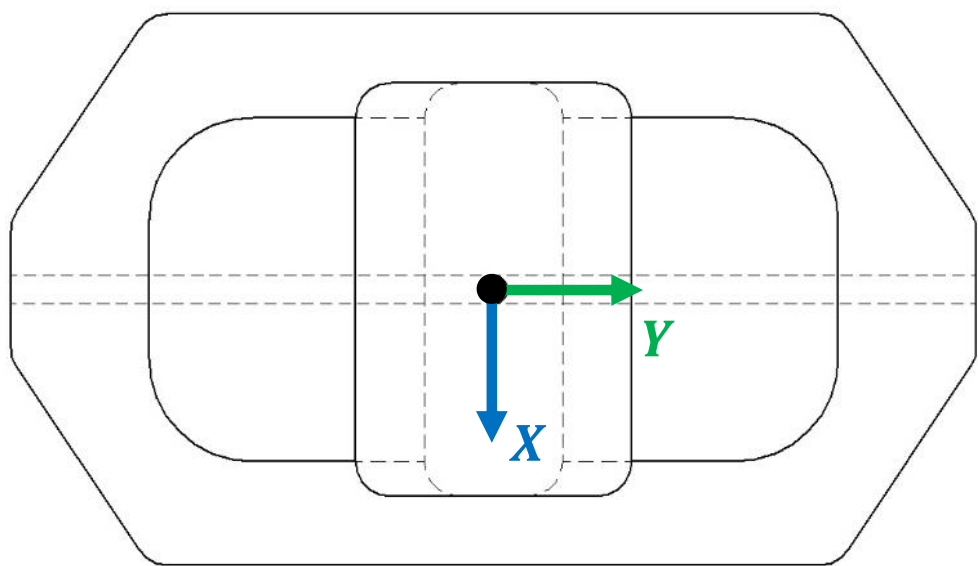
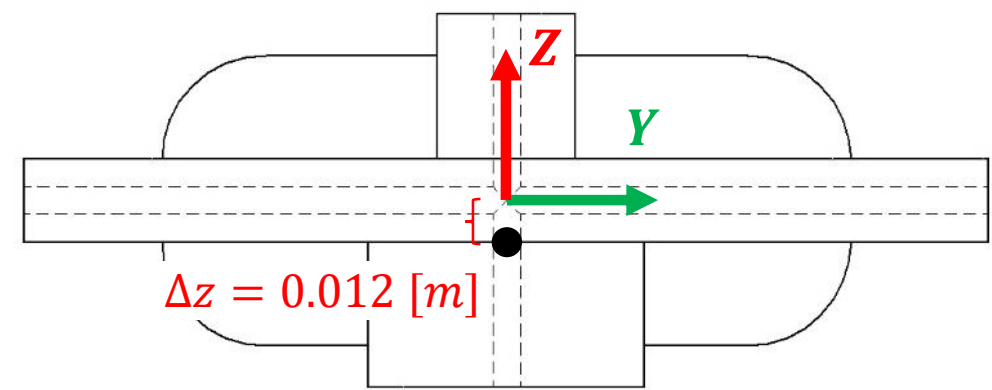
LiCAS A1 – Simplified 3D Model



LiCAS A1



LiCAS A1 Simplified 3D Model - Shoulder Pitch Structure

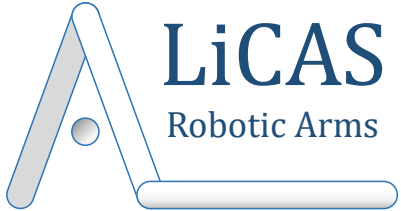
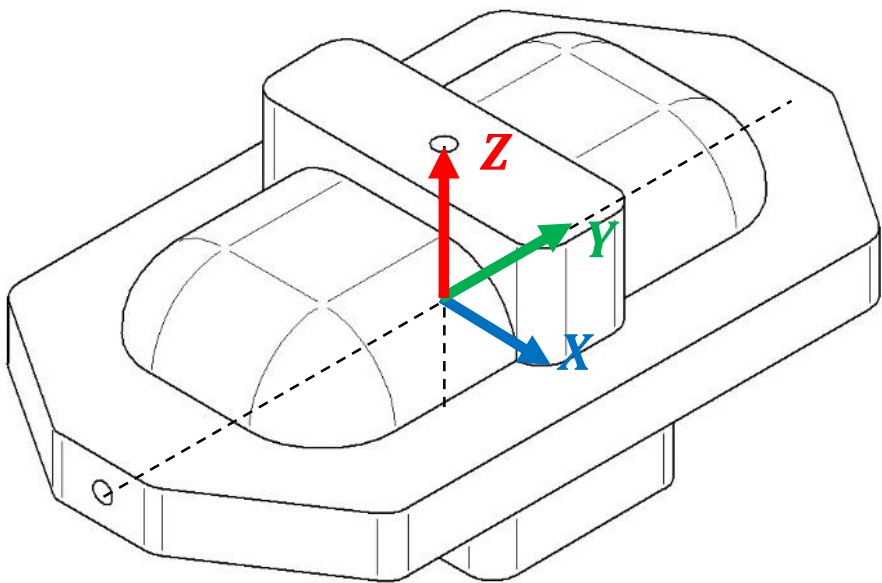


● Center of Mass (CoM)

Part name			Shoulder pitch structure		
Mass			0.639		
I_{xx}	$3.02 \cdot 10^{-3}$	I_{yy}	$8.64 \cdot 10^{-4}$	I_{zz}	$3.39 \cdot 10^{-3}$
I_{xy}	0	I_{xz}	0	I_{yz}	0
Δx_{CoM}	0	Δy_{CoM}	0	Δz_{CoM}	$-1.20 \cdot 10^{-2}$

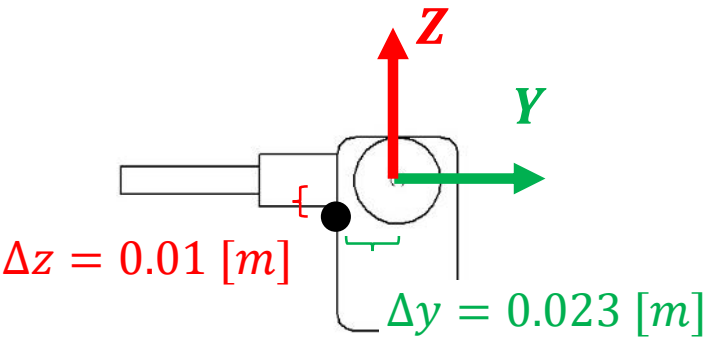
Mass: [kg] Inertia: [$\text{kg} \cdot \text{m}^2$] CoM position: [m]

Moments of inertia are referred to the CoM



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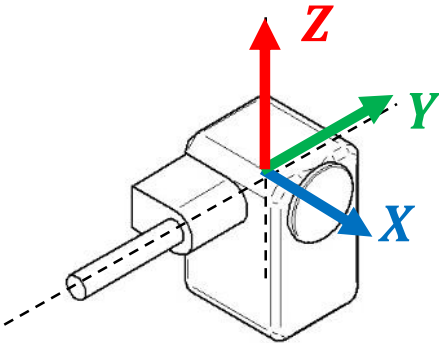
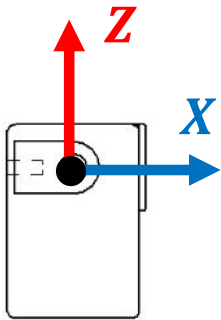
LiCAS A1 Simplified 3D Model - Shoulder Roll Structure



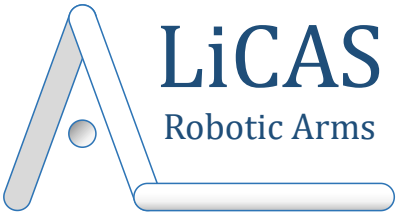
Part name			Shoulder roll structure		
Mass			0.233		
I_{xx}	$3.68 \cdot 10^{-4}$	I_{yy}	$1.16 \cdot 10^{-4}$	I_{zz}	$3.76 \cdot 10^{-4}$
I_{xy}	$-7.57 \cdot 10^{-6}$	I_{xz}	$5.38 \cdot 10^{-6}$	I_{yz}	$-4.80 \cdot 10^{-5}$
Δx_{CoM}	0	Δy_{CoM}	$2.36 \cdot 10^{-2}$	Δz_{CoM}	$9.46 \cdot 10^{-3}$

Mass: [kg] Inertia: [kg · m²] CoM position: [m]

Moments of inertia are referred to the CoM

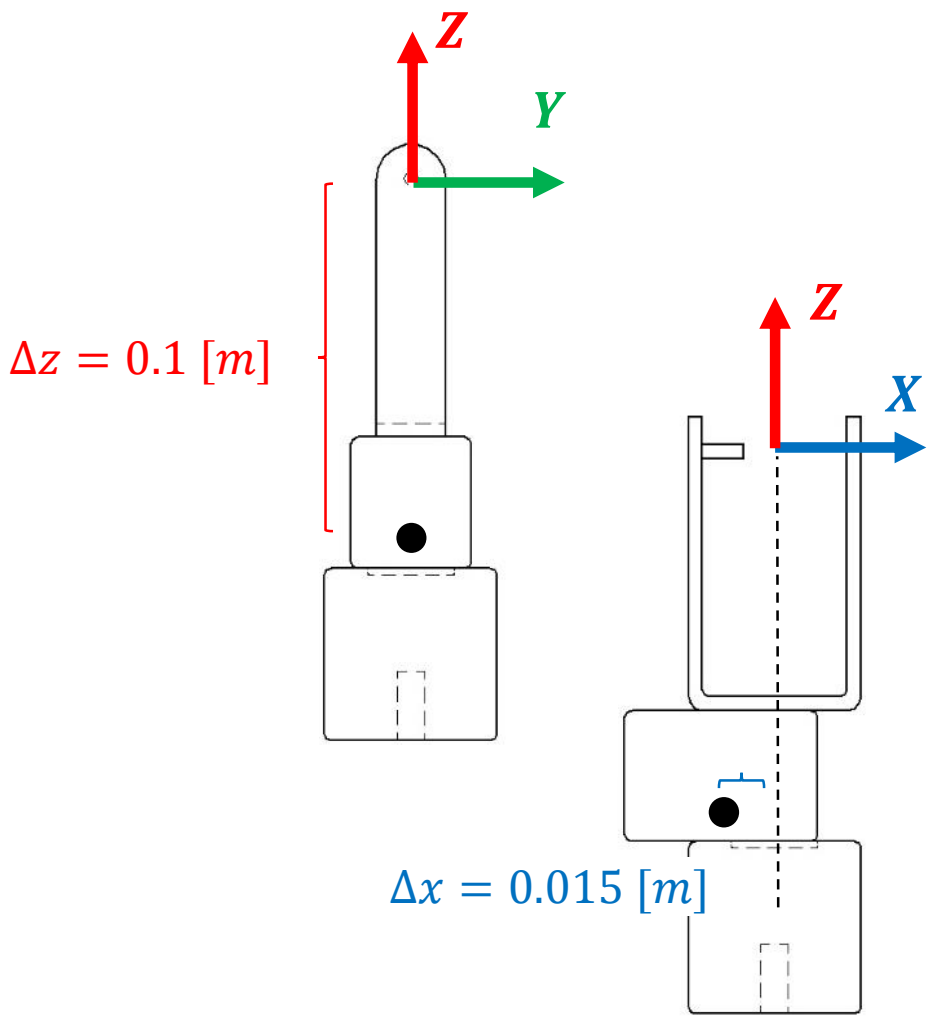


- Center of Mass (CoM)



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LiCAS A1 Simplified 3D Model - Shoulder Yaw Structure

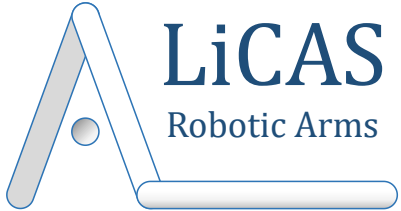
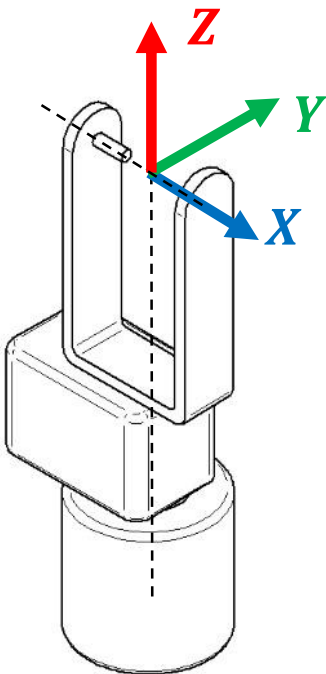


● Center of Mass (CoM)

Part name			Shoulder yaw structure		
Mass			0.246		
I_{xx}	$3.49 \cdot 10^{-4}$	I_{yy}	$4.46 \cdot 10^{-4}$	I_{zz}	$1.49 \cdot 10^{-4}$
I_{xy}	0	I_{xz}	$-3.79 \cdot 10^{-5}$	I_{yz}	0
Δx_{CoM}	$-1.50 \cdot 10^{-2}$	Δy_{CoM}	0	Δz_{CoM}	$-1.00 \cdot 10^{-1}$

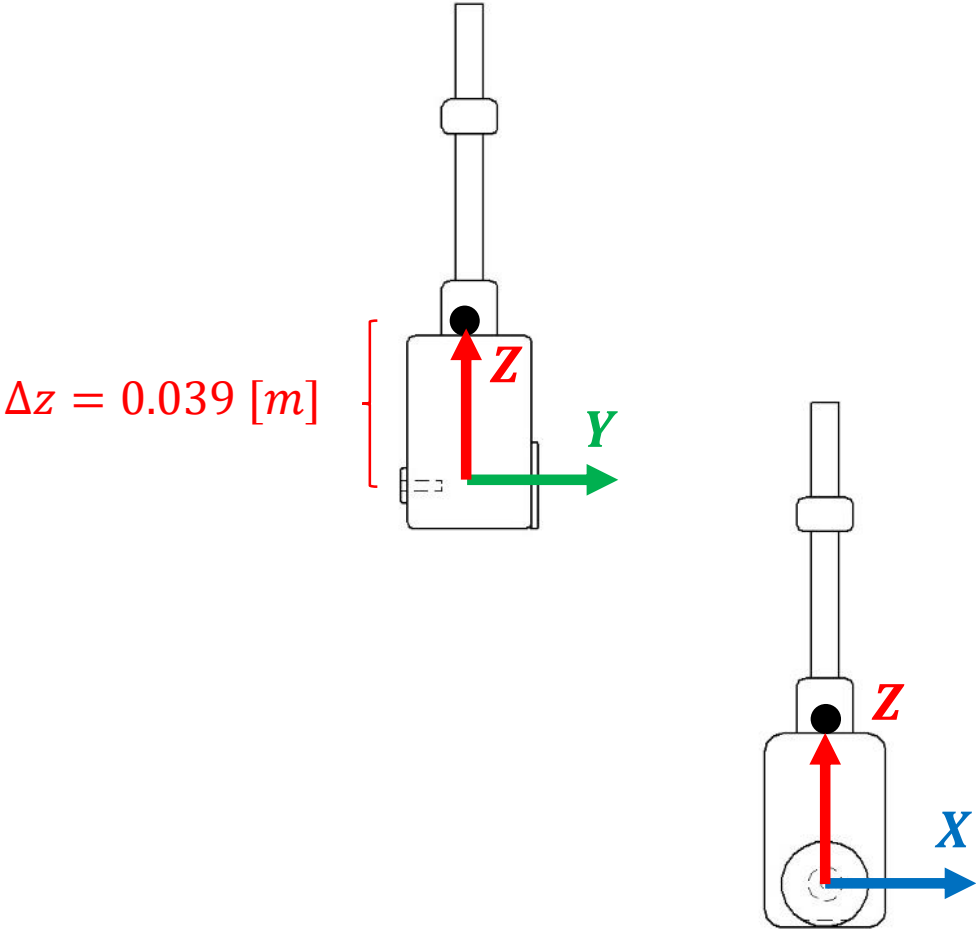
Mass: [kg] Inertia: [kg · m²] CoM position: [m]

Moments of inertia are referred to the CoM



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LiCAS A1 Simplified 3D Model - Elbow Pitch Structure

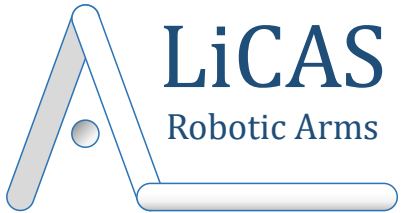
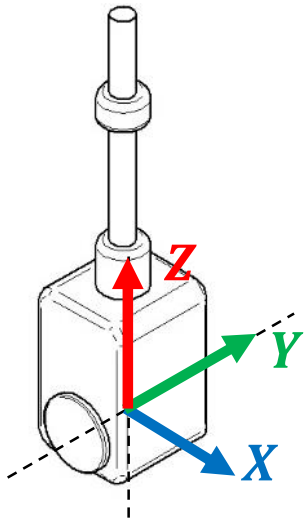


● Center of Mass (CoM)

Part name			Elbow pitch structure		
Mass			0.214		
I_{xx}	$4.21 \cdot 10^{-4}$	I_{yy}	$4.26 \cdot 10^{-4}$	I_{zz}	$5.01 \cdot 10^{-5}$
I_{xy}	$-7.81 \cdot 10^{-7}$	I_{xz}	$-4.06 \cdot 10^{-5}$	I_{yz}	$1.80 \cdot 10^{-5}$
Δx_{CoM}	0	Δy_{CoM}	0	Δz_{CoM}	$3.90 \cdot 10^{-2}$

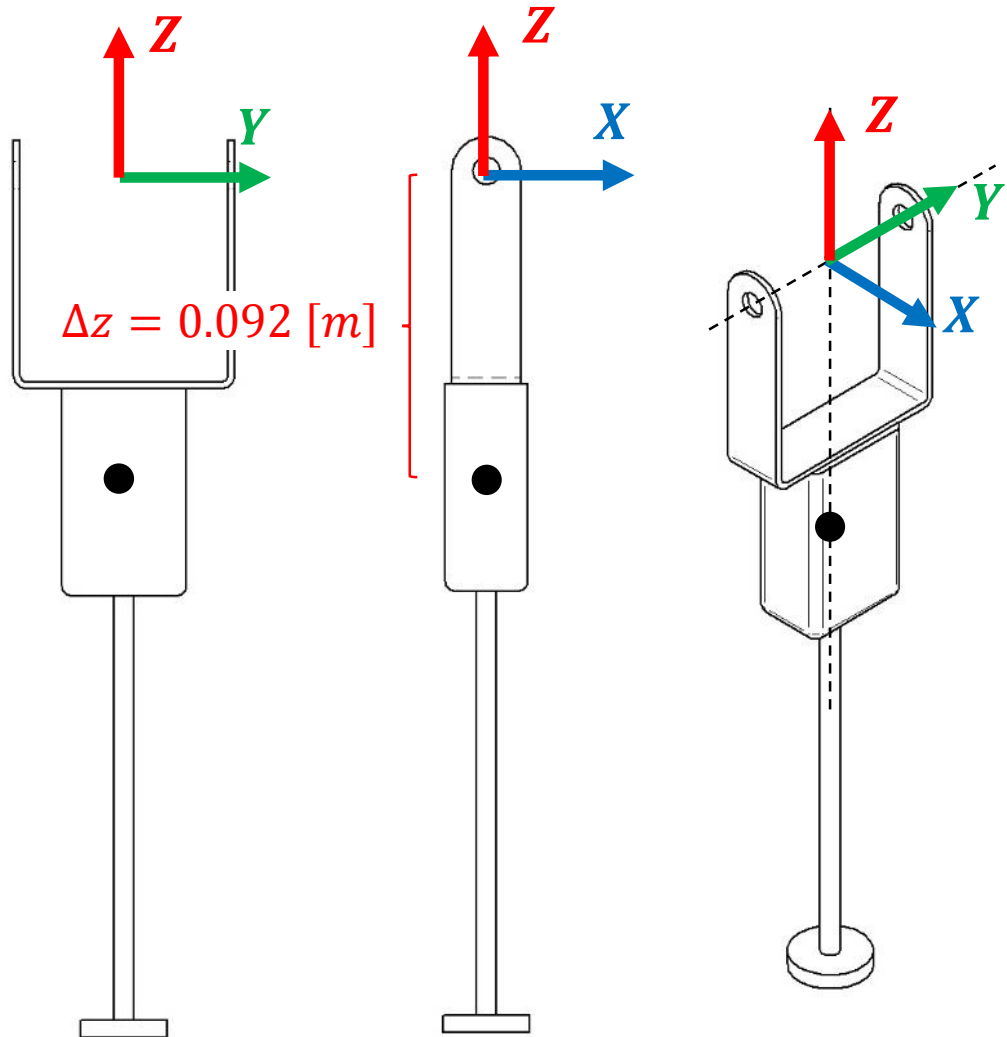
Mass: [kg] Inertia: [kg · m²] CoM position: [m]

Moments of inertia are referred to the CoM



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LiCAS A1 Simplified 3D Model – Forearm Link with Gripper Servo

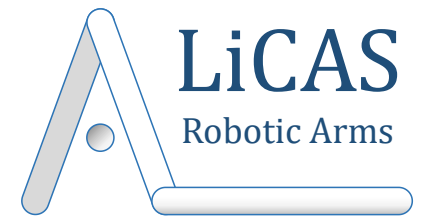


- Center of Mass (CoM)

Part name			Forearm link + gripper servo		
Mass			0.106		
I_{xx}	$3.94 \cdot 10^{-4}$	I_{yy}	$3.75 \cdot 10^{-4}$	I_{zz}	$3.05 \cdot 10^{-5}$
I_{xy}	0	I_{xz}	0	I_{yz}	$-3.50 \cdot 10^{-6}$
Δx_{CoM}	0	Δy_{CoM}	0	Δz_{CoM}	$9.20 \cdot 10^{-2}$

Mass: [kg] Inertia: [$kg \cdot m^2$] CoM position: [m]

Moments of inertia are referred to the CoM



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